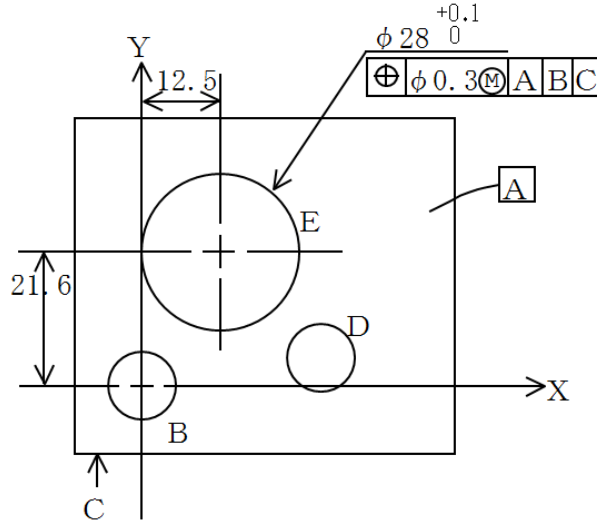


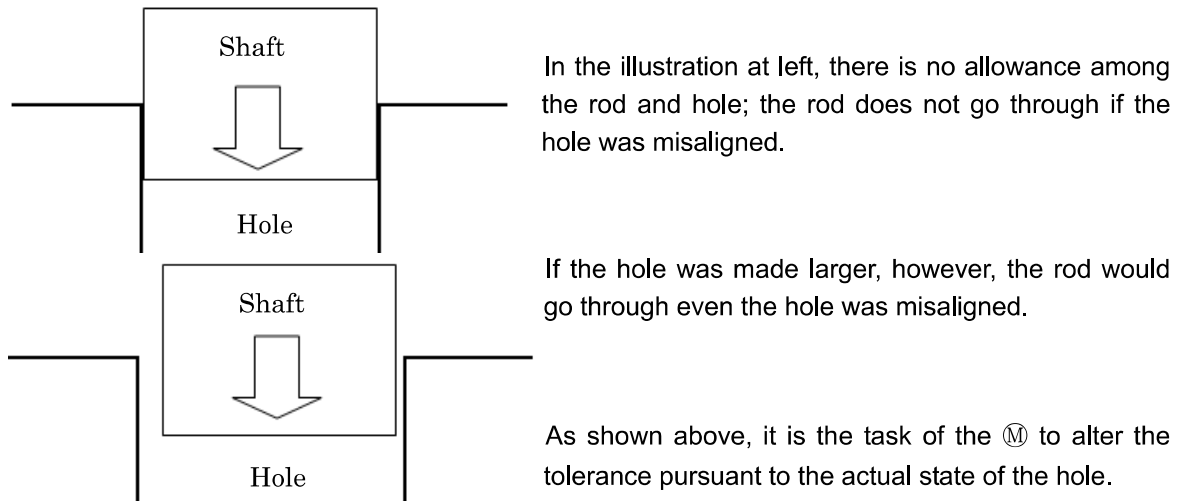
Appendix9. Setting the $\textcircled{M}$ (Maximum material principle)



Calypso is made capable of evaluating the Maximum material principle ($\textcircled{M}$) as illustrated above.

What is $\textcircled{M}$?

It is a method to set the tolerable value when evaluating the positional relation of a hole and rod.



To use the $\textcircled{M}$

To use the $\textcircled{M}$, you need to evaluate the diameter of the circle of the evaluation objective as a task. Allowance of the geometric tolerance shall be enlarged in proportion to that diameter task.

Evaluation mode of the $\textcircled{M}$

Two modes are prepared to evaluate the $\textcircled{M}$.

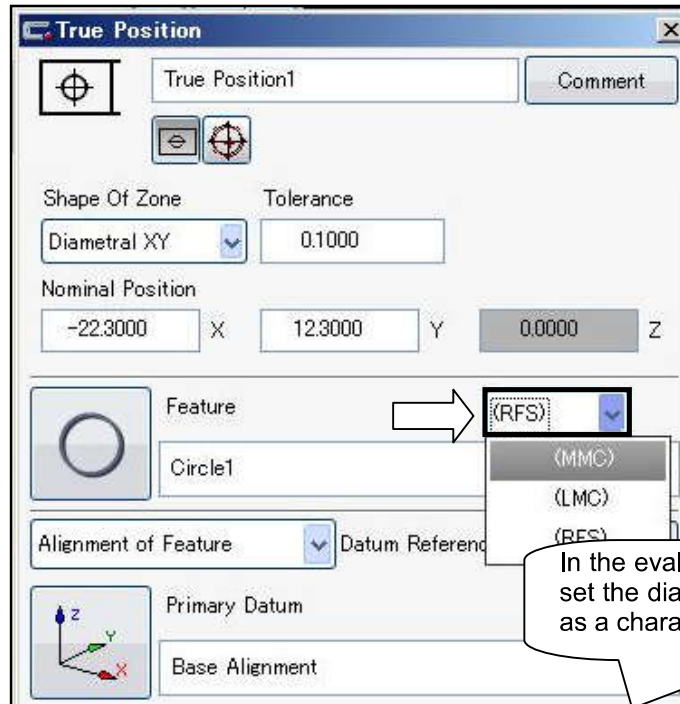
- A. If smaller than the lower limit value of the diameter, allowable tolerance shall be enlarged without limit.
- B. Allowable tolerance shall be enlarged up to the upper limit of the diameter.

Calypso has the setting A as default.

You can change the mode from the [Measurement plan editor task] in the [Preparation] menu.

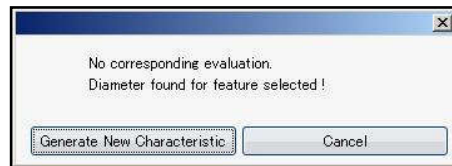
Setting procedure of the $\textcircled{M}$

- ① Upon evaluated the Positional tolerance, a menu called [RFS] appears at the right of selected element. Select [MMC] in that menu.

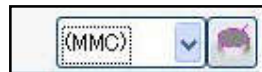


In the evaluation of the MMC, you need to set the diameter of the circle having the M as a characteristic.

- ② Following message shall appear if the diameter of the evaluation element was not set as a characteristic. In such a case, left click [Create New Characteristic].



- ③ As an icon appears at the right of the [MMC], you can set the diameter characteristic of the Reference circle. Close the window by the OK button at last to complete evaluation.



Appendix10. Setting the $\textcircled{P}$ (Protrusion tolerance)

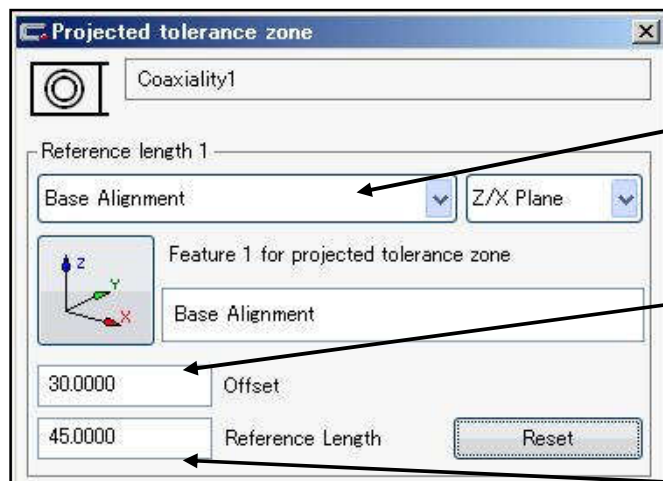
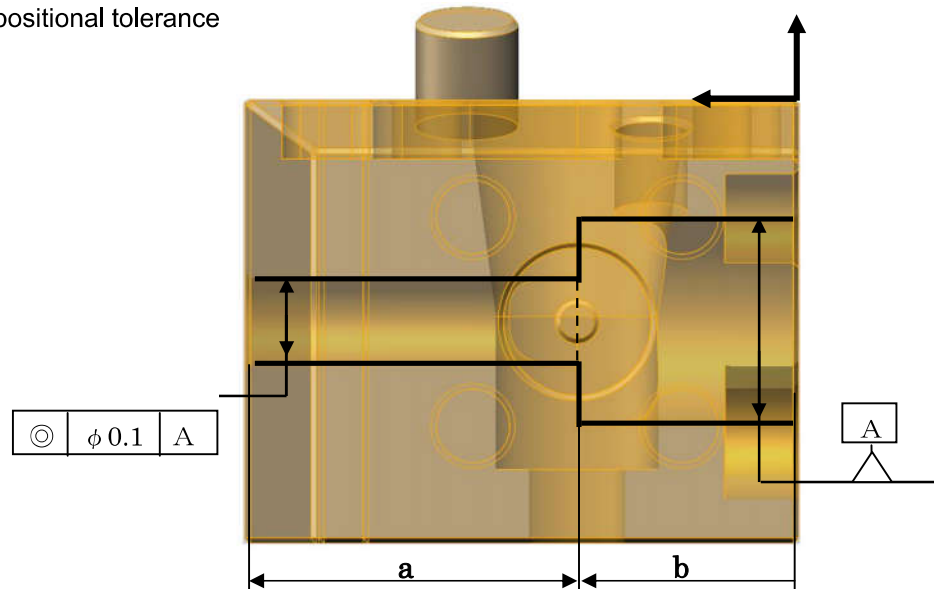
This is a function used to evaluate the geometric tolerance of the rod (Cylinder, cone, 3D line).
Items applicable to set shall alter according to the type of the geometric tolerance to evaluate.

Setting 1. Length of the rod to evaluate (Length a in the photo)

Parallelism, squareness, angularity, coaxiality, positional tolerance.

Setting 2. Position of the rod to evaluate (Length b in the photo)

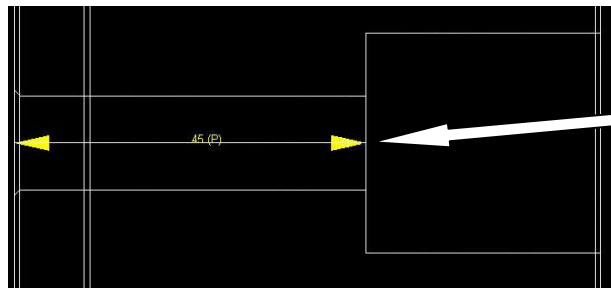
Coaxiality, positional tolerance



① Select the alignment

② Coordinate value that the evaluation cylinder starts (b).

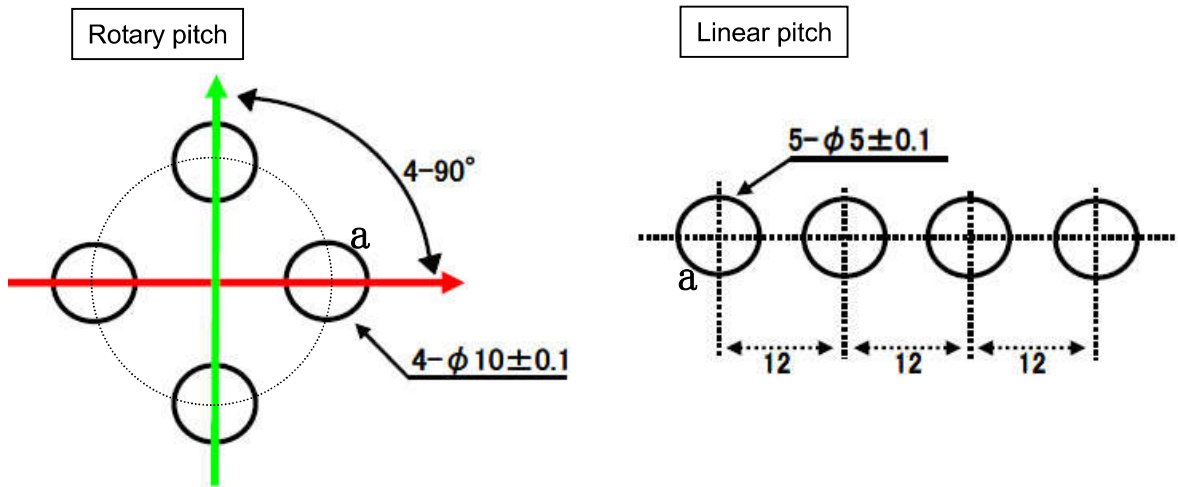
③ Length of the evaluation cylinder (a).



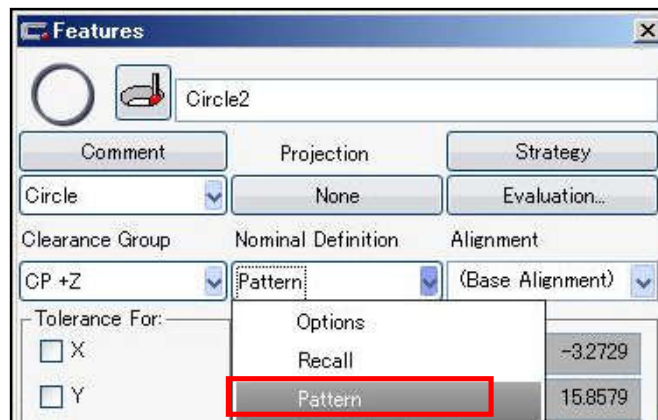
You can monitor the evaluation range on the CAD display.

Appendix11. Set Pattern

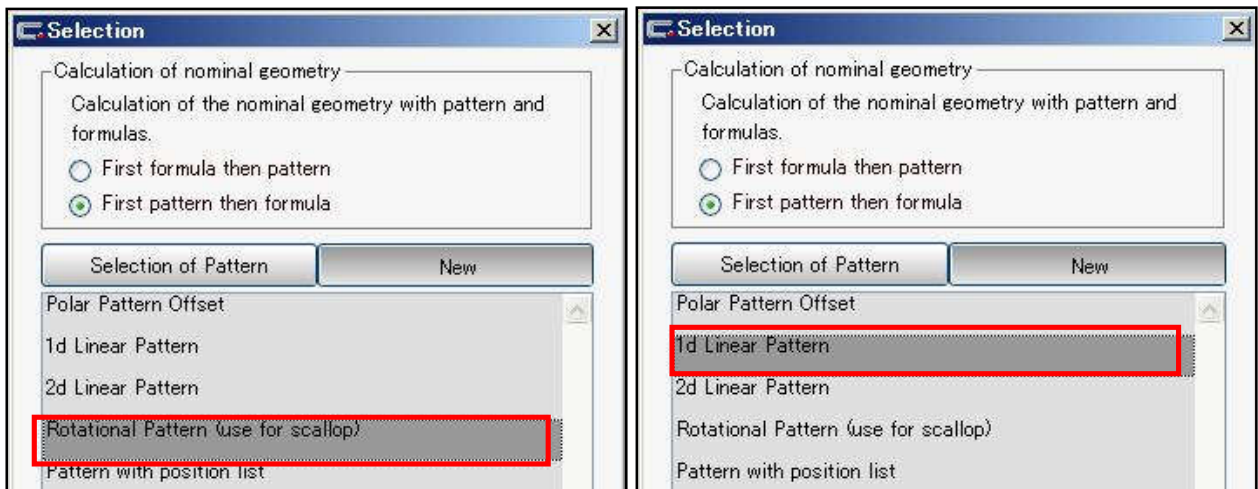
In case where elements were positioned according to a certain rule, you can simply create plural elements by using the Set Pattern.



- ① Measure the hole a. (On provisionally putting it as the circle 2.)
- ② Open the circle 2 and select [Pattern] from the menu beneath the [Nominal Definition].



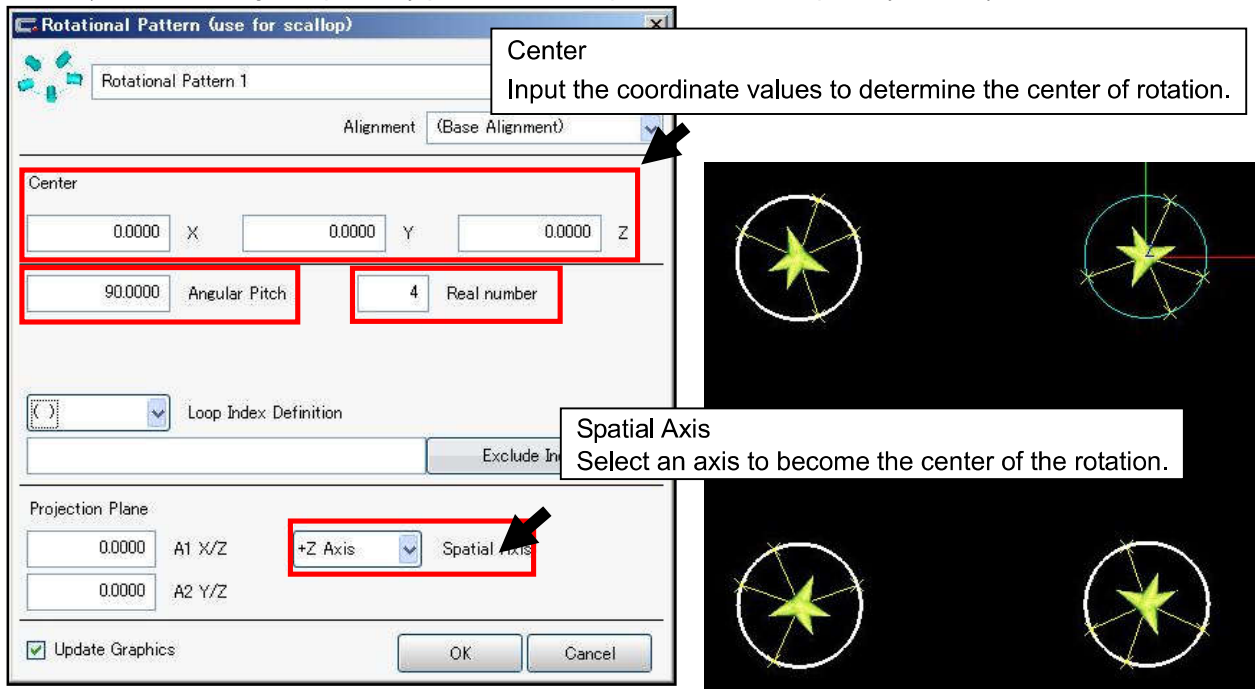
- ③ As the Selection menu window is displayed, select an optional pattern and left click **OK** button.



④[Rotational Pattern]

As the Rotational Pattern display appears, input 90 in the “Angular Pitch” and 4 in the “Real number”.
It shall mean the setting that there are four holes per every 90°.

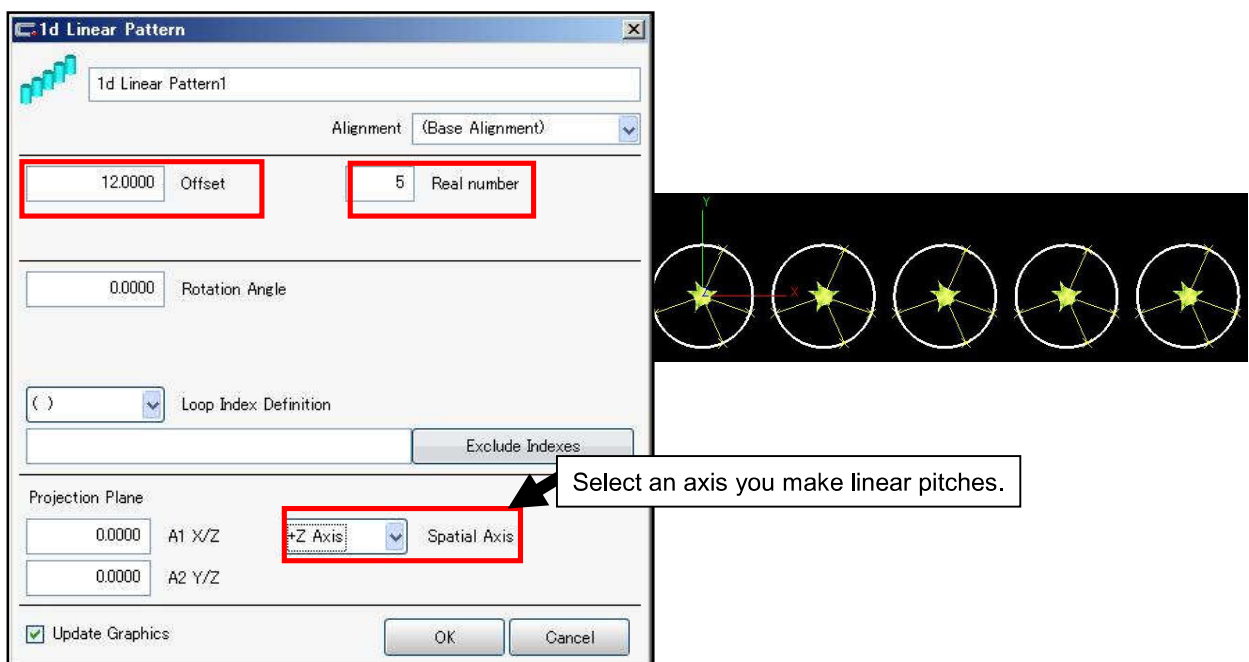
(You can change it optionally pursuant to the profile of the workpiece you use.)



[Linear Pattern]

As the Linear Pattern display appears, input 12 in the “Offset” and 5 in the “Real number”.

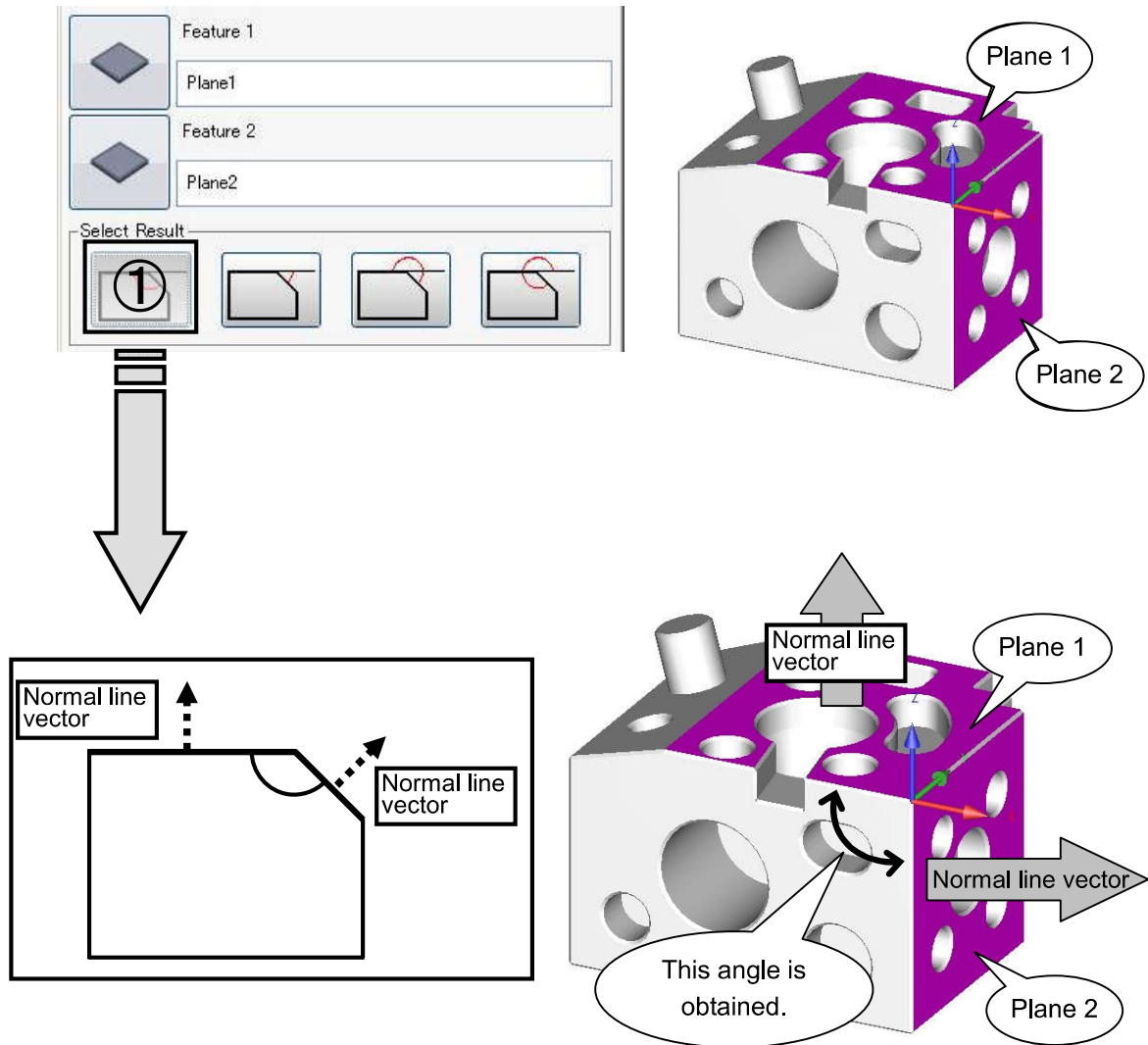
It shall mean the setting that there are five holes with the interval 12mm.



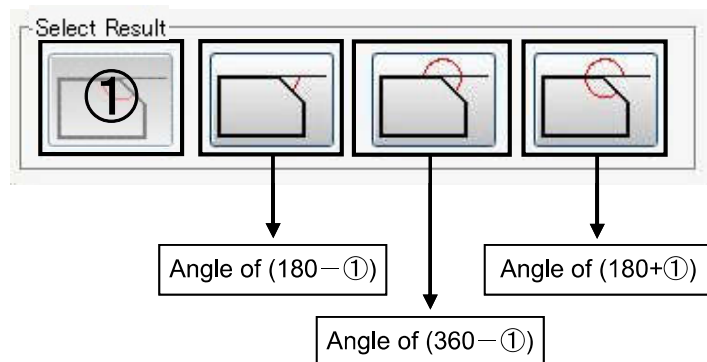
⑤Monitor the CAD display and left click OK button if there was no problem.

Appendix12. Outputting the angle

If the angle between planes was 90 degree, you can refer to the following to see the position that an angle is output.

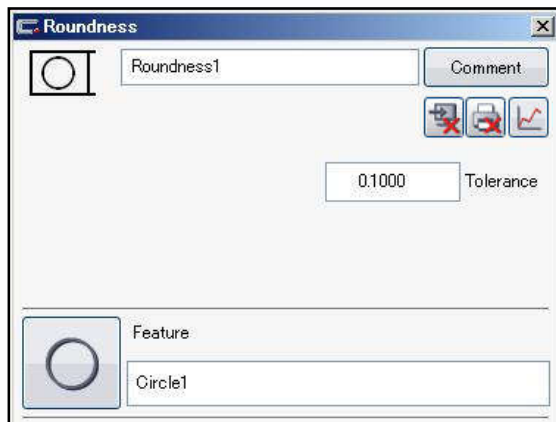


他の角度出力は？



Appendix13. Display of the profile plot

There are cases that the Profile plot display applies in the geometric tolerance roundness and flatness and so on. Open the Characteristic window to do respective setting on the Profit plot display.



Displays the profile plot in the display upon completed measurement.

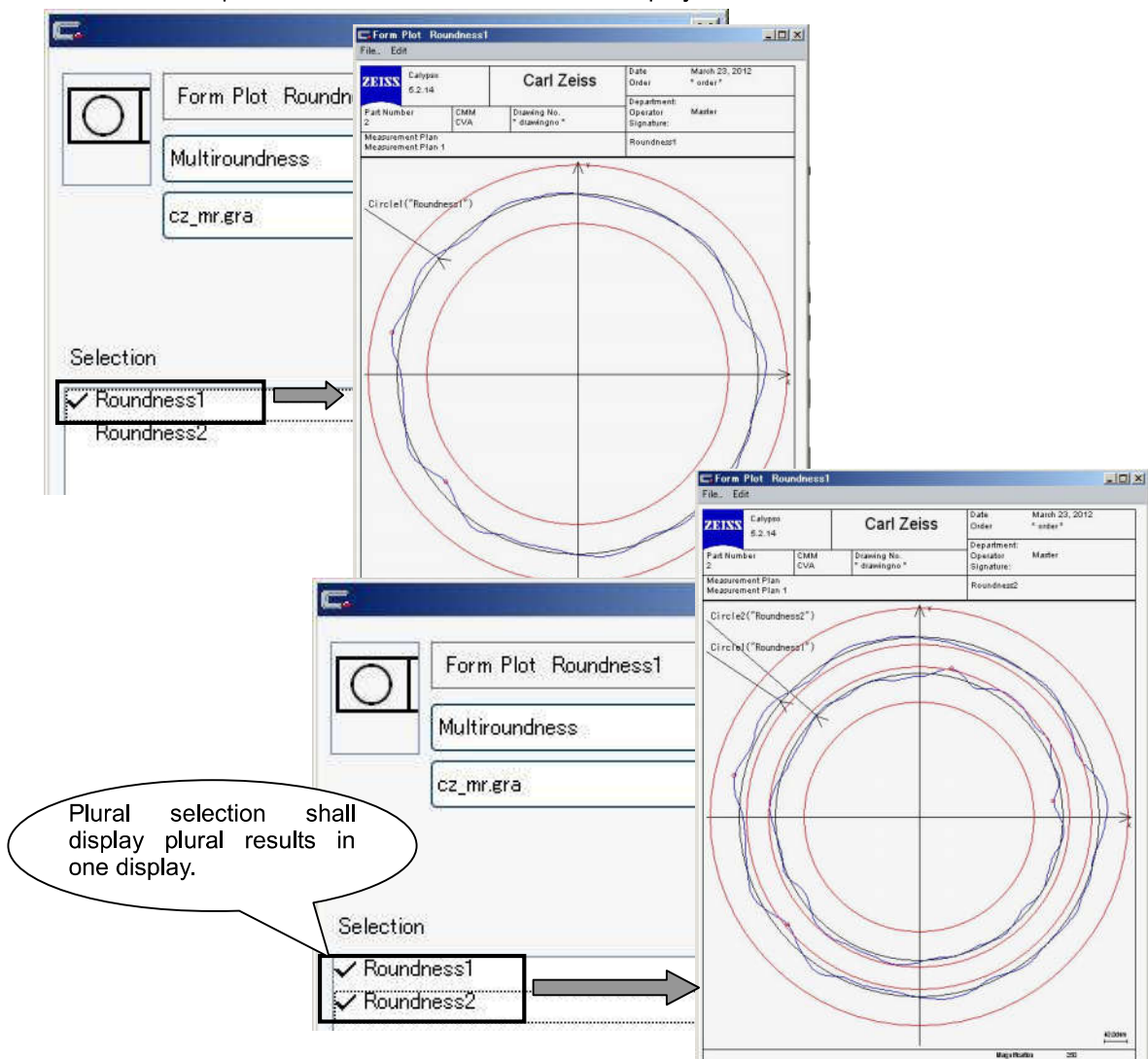


Prints out the Profile plot in the display upon completed measurement.

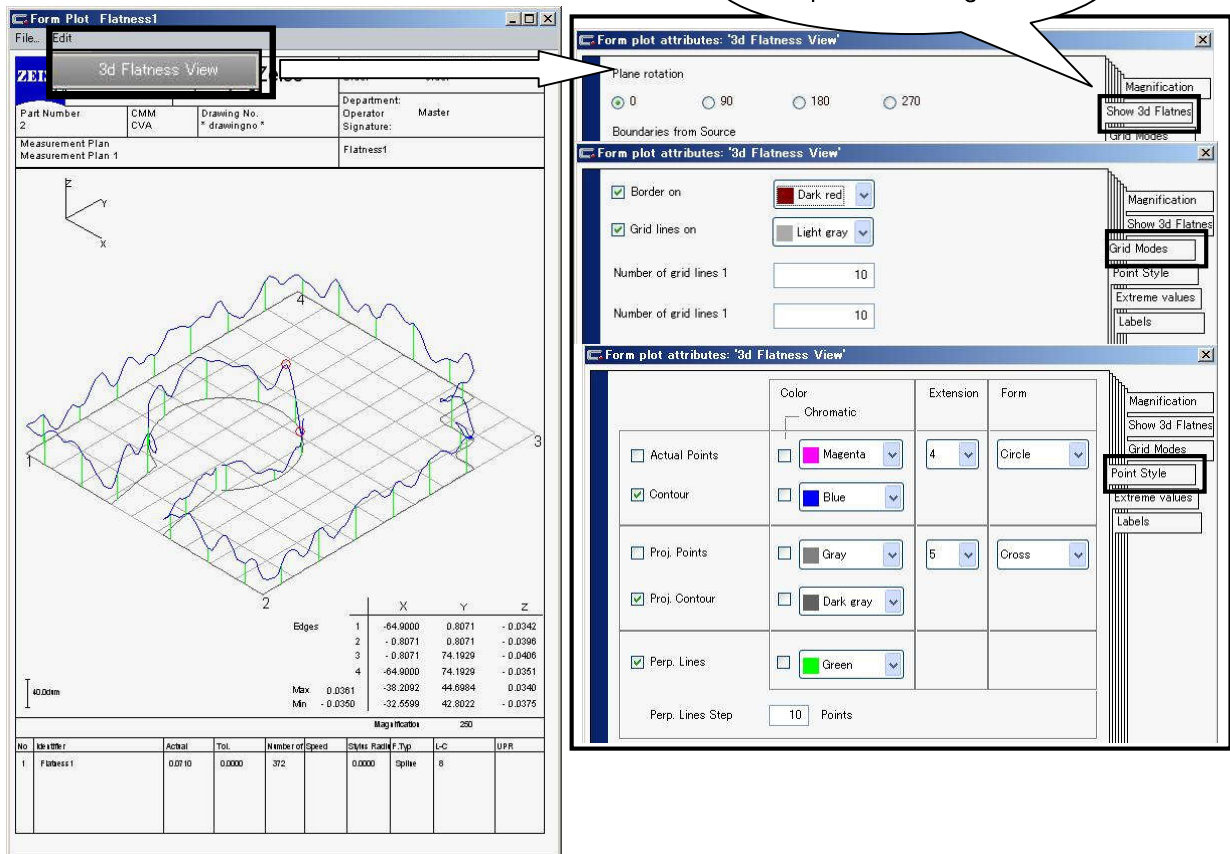
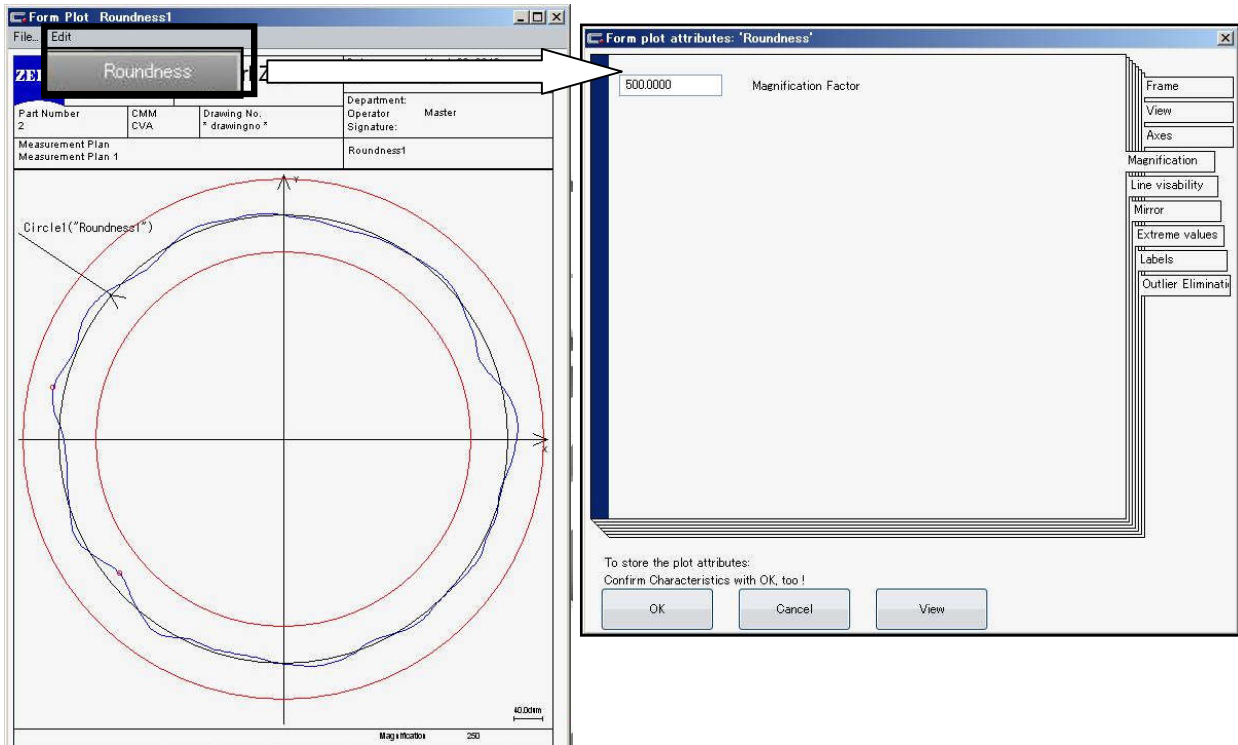


Displays the profile plot in the display.

Left click the Profile plot icon to set the result to display.



Use the [Edit] menu to do respective setting of the profile plot display.



Appendix14. Transmission of the measurement plan

On the interchange of data and/or discussion on measurement with the customer, your counterpart can monitor the completely same display as yours if the measurement plan itself was transmitted adding to the measurement result.

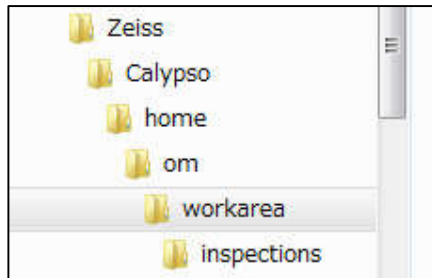
Let us introduce here that procedure.

- ① Open a plan to be the objective and overwrite and save it.



- ② Next, close (or minimize) the Calypso to display the desktop.
- ③ Open the My computer by left clicking double the mouse. Follow folders one by one as per the procedure below.

Drive C⇒Program Files(x86)⇒Zeiss⇒Calypso⇒home⇒om⇒workarea⇒inspections

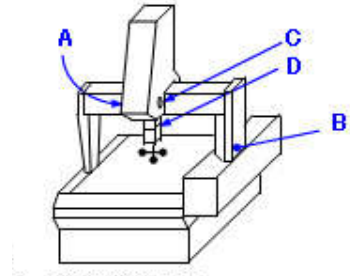


- ④ In the end you'll see a folder having the name same as the saved measurement plan.
- ⑤ Right click the mouse on the target folder and select [Send (N)] – [Compressed (zip format) folder] from the menu.
- ⑥ As a compressed file having the same name is created, move that file in the CD-R or USB memory and the like.

Appendix15. Trouble shooting

1.Drive is not available

1. Does the supplied air have the prescribed pressure (not too low)?
2. Isn't the emergency button pressed? The emergency can be released by turning the button right or left.
3. Does any one of the X, Y, or Z axes go over the limiter?
Press "A" part for the X-axis and press back side of "B" part for the Y-axis.
Grab and lift "C" part for the Z-axis.



2.Environment backup

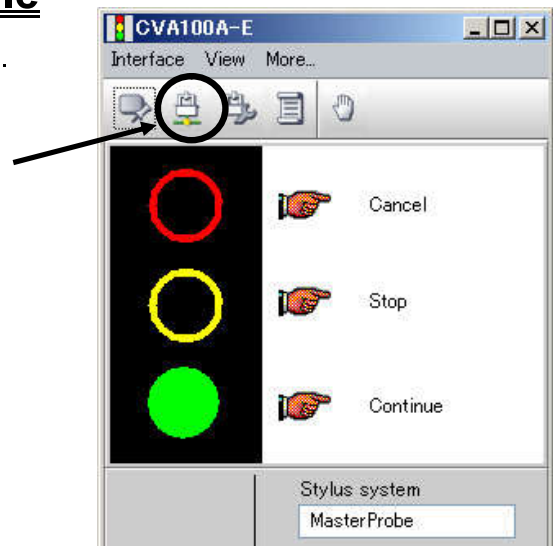
You can backup the measurement environment and measurement plan if you copied the following folder.

「C:¥Program Files(x86)¥Program Files¥Zeiss¥Calypso¥home¥om¥config」 ⇒Machine environment

「C:¥Program Files(x86)¥Zeiss¥Calypso¥home¥om¥workarea¥inspections」 ⇒Measurement plan

3.When the operation is unstable

1. When a trouble such as probe collision is occurred.
The signal turns in red. Click the green signal.
The connection can be initialized by clicking the connector icon in the Signal window. If the collision is too intense under a high-speed operation or other conditions, the following error possibly is displayed.
"The measurement machine lost the home position. Establish new connection from the beginning".
In this case, restart the measurement machine.



2. The screen does not change (Control panel CNC lamp lights up)
The measurement machine is set to the CNC (automatic drive) mode.
It can be stopped with the ESC key or the Emergency Stop button.

3. Other unstable conditions (general counter measurement)

Connection initialization

Click the Connector icon in the Signal window.

Calypso restart

If the operation of the measurement machine cannot go back to normal even doing a re-connection, restart the Calypso.

Normal termination

Once terminate the Calypso, and then restart the machine.

Forced termination (Ctrl+Shift+Esc)

Open the task manager and terminate the task of the "Calypso User Desk."

Restart of measurement machine

If the error still remains after restarting the Calypso, restart the measurement machine main unit.

* Before restarting the machine, once terminate the measurement machine according to the procedure described on page 5.

(If the machine does not go back to normal even doing the above operation, please contact us)

4. Stylus naming convention

For the stylus and stylus system names, do not use special characters.

Only underbars (_) can be used as a separator.

5. Daily check

1. Traveling surface of the measurement machine

Perform regular cleaning using alcohol.

2. Stylus check by calibration

Calibrate the stylus and check the standard deviation.

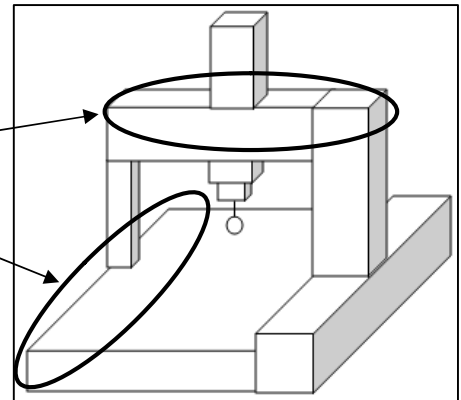
Standard deviation shall become the target numeric value to grasp the state of the stylus.

Check and ensure that the deviation value remains the same.

3. Air-filter exchange

Check and replace the air-filter of the measurement machine regularly.

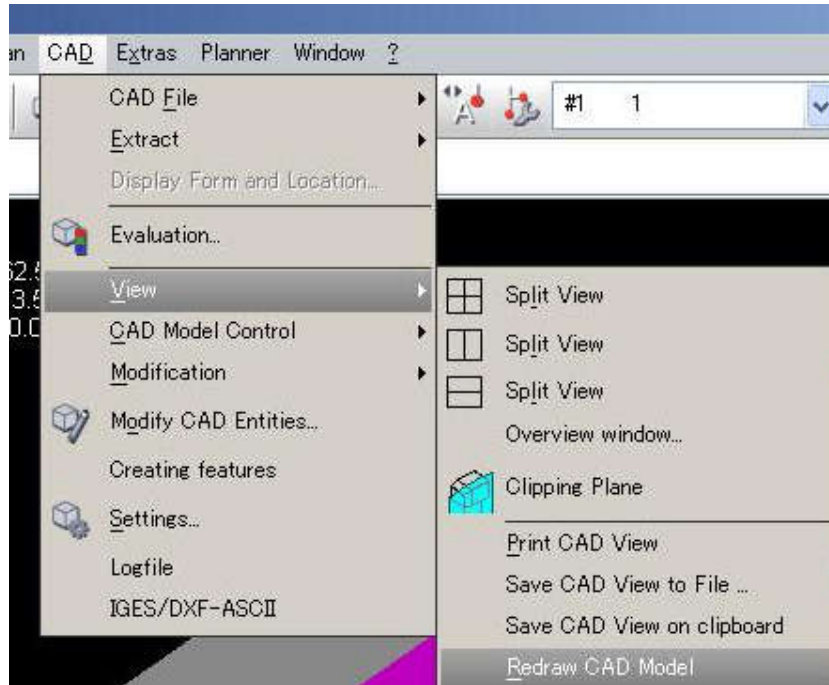
Depending on the model, the air-filter is located at the back or the side of the measurement machine.



6.CAD screen cannot be drawn normally

In some cases, the CAD screen may not be drawn normally.

In that case, from the menu bar, click "CAD", "View", and "Redraw CAD model."



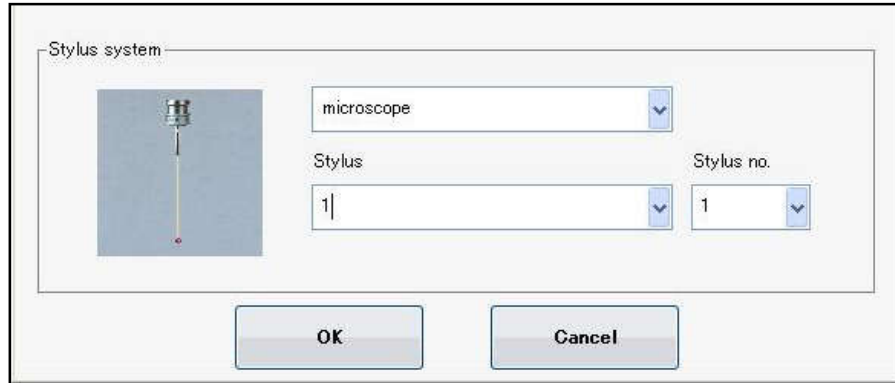
7.Setting the Windows theme

Windows theme is set as the Windows classic at shipment.

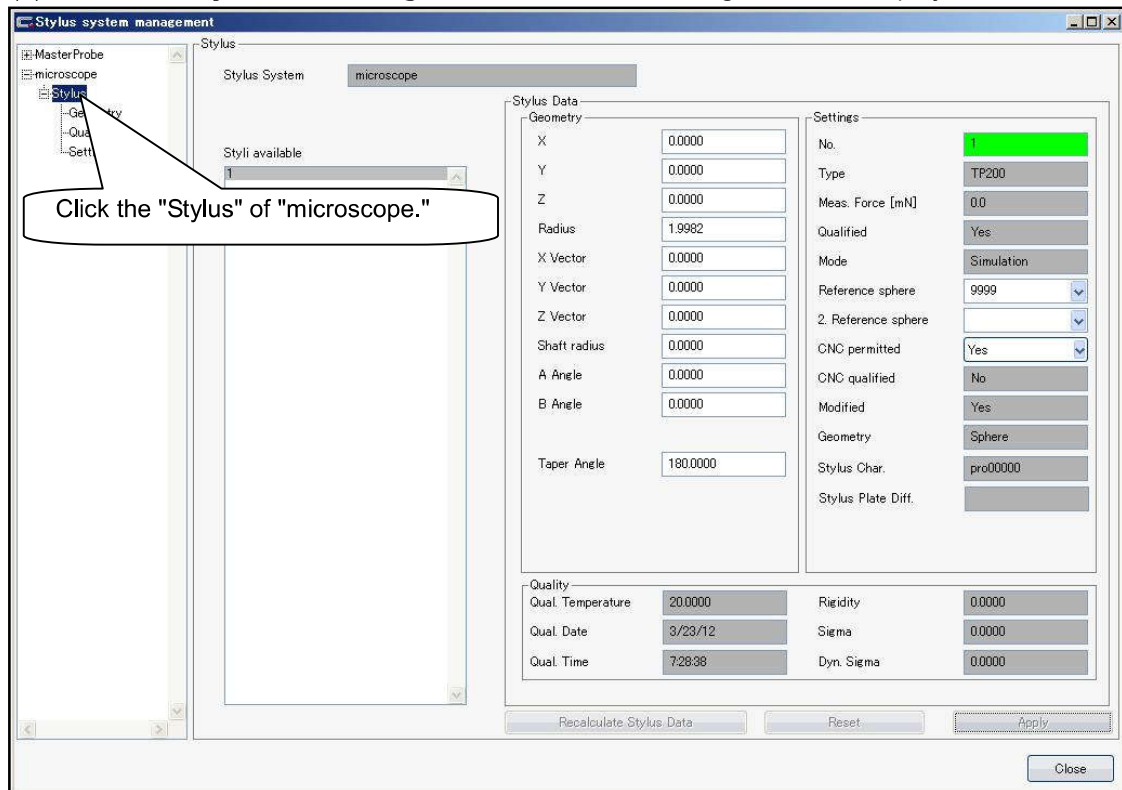
Do not change the setting as it might hamper correct operation of the Calypso.

Appendix16. Microscope

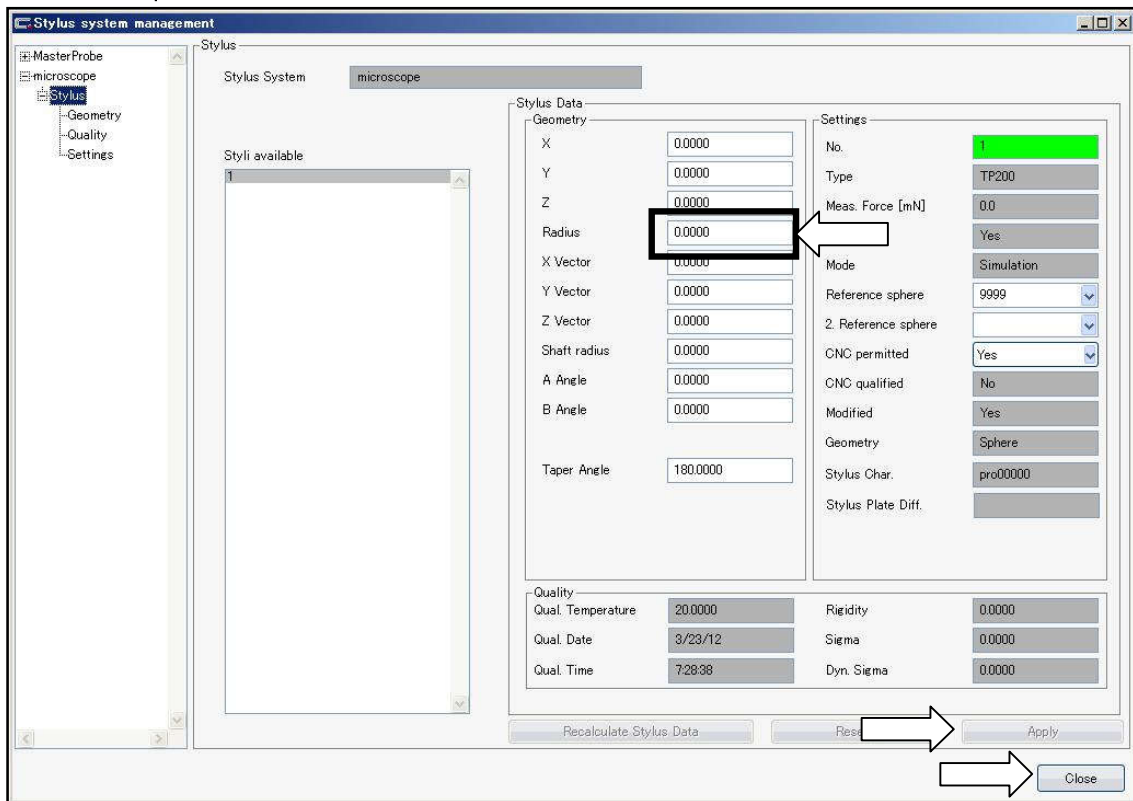
1. Click "Create New Measurement Plan icon", input a name, and click "OK."
2. Register the microscope. Perform the following operation while maintaining the electron probe attached.
 - (1) Click "Stylus Icon" in the Prerequisites list and perform "Ref. Sphere Position" with the Master Probe (Refer to Stylus Calibration).
 - (2) Click "New Stylus System Definition" icon, input "microscope" for the Stylus System name and "1" for the 1st stylus name, and click "OK."



- (3) Perform **stylus calibration** with the Master Probe attached.
- (4) Next, click "**Stylus Data Management**" icon. The following window is displayed.



- (5) Click the numerical portion of the stylus radius, "**Radius**", and change the numerical value in the input box to "0."



- (6) Click "**Apply**" and "**Close**" to go back to Stylus Calibration window. Click "OK" again.

Then, the calibration for microscope is completed.

The above step is no longer necessary once it is completed for using a microscope for the first time.

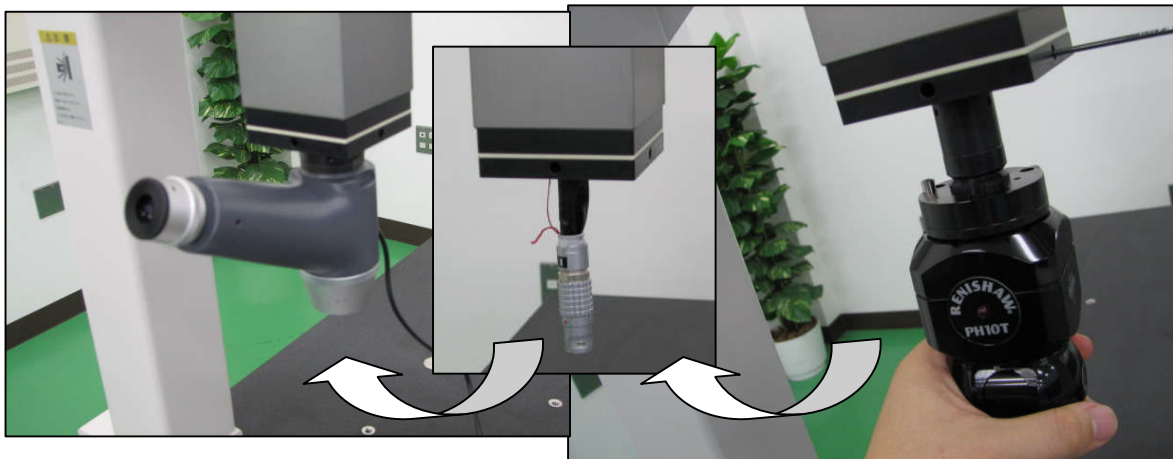
3. Mount the microscope.

Keep the measurement machine turned off, remove the head, and mount a dummy connector only.

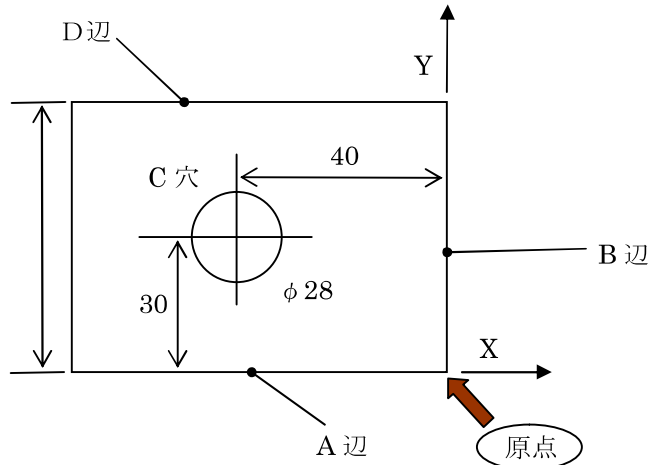
Turn the measurement machine power on to activate Calypso.

(The microscope is not yet mounted at this time)

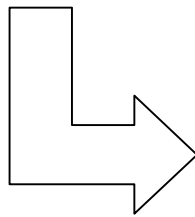
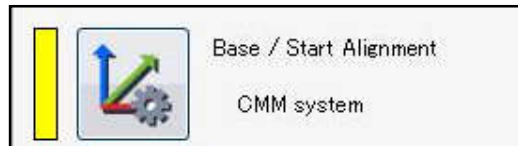
Mount the microscope after the measurement machine moves to the home position.



4. Specify Base Alignment. An example as following figure is described here.



(1) Open the Setup screen for Base Alignment



The 'Base Alignment' dialog box is shown. It contains the following fields and options:

- Measurement Plan 2-BS** (with a 'Special' button)
- Measurement Plan 2** (with a 'Comment' button)
- Spatial Rotation** (with a dropdown menu)
- Planar Rotation** (with a dropdown menu)
- X Origin** (with a text input field)
- Y Origin** (with a text input field)
- Z Origin** (with a text input field)
- ☐ **Manual Alignment**
- Execute During Run As:**
 - ☒ **Automatic Measurement**
 - ☒ **Set base alignment to zero**
- Execute Manual Run Now** (with a button icon)
- OK** and **Reset** buttons at the bottom.